

LECTURE 3

Pre-training I: data, budget and the run

Gleb Molodtsov · 21 September 2026



Architecture is far from everything

$\min L(\text{data}, \text{architecture}, \text{optim.}, N, D)$

$\text{s.t. } C \approx 6ND$

- **Four knobs come first.** Today we turn all four
- **Not only pre-training.** The same knobs run mid-training, continual PT and domain adaptation

NOT TODAY

GPU, memory, parallelism — lecture 4

SFT, RLHF, DPO — lecture 5

ON THE BOARD ALL LECTURE

This equation. Every slide turns one of its arguments.



Five decisions, in the order you actually make them



01

Budget

where the GPU-hours really go

02

Target

the model you want fixes the recipe

03

Data

the web, the pipeline, three surprises

04

Optimization

AdamW, Muon, schedules, transfer

05

Scale

6ND, Chinchilla, phases of a run



01

• Budget

Where a million GPU-hours actually goes



You have 1,000,000 GPU-hours. Split it.

Eight lines. Write your percentages down — we check them in a minute.



___ data pipeline

___ ablations

___ HP search

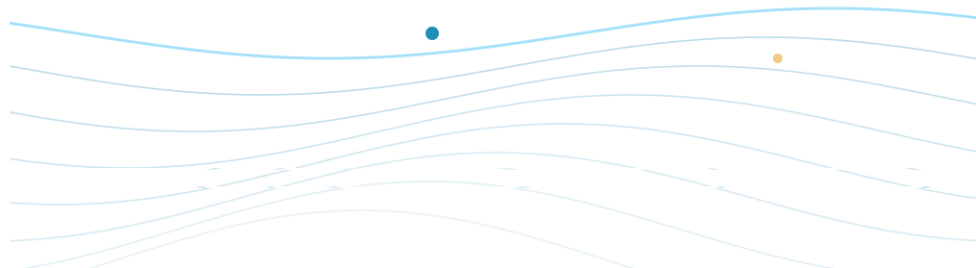
___ final run

___ restarts

___ evaluation

___ long-context

___ mid-training



The final run is a minority of the bill

Measured, not guessed: three companies, three disclosures, one answer.

OpenAI
2024 · \$5.0B R&D compute **9.6%** everything else — experiments, synthetic data, unreleased models, basic research

Z.ai
H2 2024 – H1 2025 · \$216M **12.3%**

MiniMax
Q4 2024 – Q3 2025 · \$141M **22.6%**

share of R&D compute spending that went into the final training runs of released models

Epoch AI, «Final training runs account for a minority of R&D compute spending», 23.03.2026 · OpenAI via The Information, Mini Max and Z.ai from their January 2026 IPO prospectuses · this is spending, not FLOP: R&D runs at lower MFU than a final run



Experience should make the next run cheaper

It does — but less than anyone expects.

Transfers almost fully

- training framework and kernels
- cluster and dataloader engineering
- eval harness and its dashboards
- team judgement about what to try next

Transfers partially

- pipeline code, filters get re-tuned
- scaling-law method, not its constants
- μ P transfer, one parameterisation only
- raw corpus, re-extracted per tokenizer

Does not transfer

- the FLOPs of the run itself
- architecture ablations after a reshape
- mixture weights for a new target
- anything tied to the old context

- **The prediction:** labs behind the frontier copy what works and experiment less
- **The test:** MiniMax fits at 22.6%, Z.ai does not at 12.3% — three points, no conclusion



02

• Target

The model you are shipping fixes almost everything else



Four targets, four different recipes

Flash / edge

latency, \$/token

- small dense or thin MoE
- distil from your own Pro
- over-train past optimum

Pro / frontier

capability ceiling

- large sparse MoE
- train from scratch
- mid-training for reasoning

Small agentic

tool loops, long context

- long-context phase required
- code and trajectories in the mix
- cheap KV beats raw quality

Knowledge-heavy

facts per parameter

- total params over active
- unique tokens over epochs
- domain corpus with replay

MoE-or-dense is one row of this table, not a separate question.



The cheapest answer is usually not to train

prompting → RAG → SFT of an open model → continual PT → pre-training from scratch

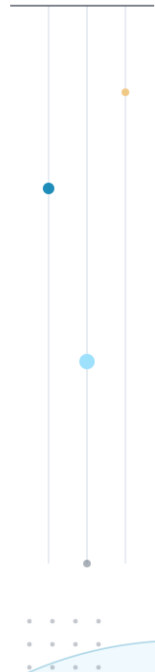
each step up costs roughly 10× more

YOU MAY STILL FINE-TUNE, MID-TRAIN OR ADAPT IF

- **You own a corpus** nobody else has trained on
- **Your domain** has no competent general model
- **Your language** is under-served by open models

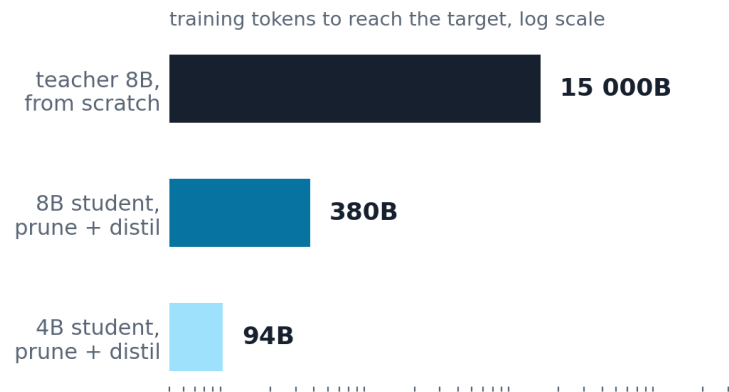
BAD REASONS, HEARD EVERY QUARTER

- «We have idle GPUs»
- «Everyone is doing it»
- «AI is the future»



A warm start buys tokens

- **40× fewer tokens** for an 8B student — 380B against 15T
- **150× fewer** for a 4B student, and it beats that size from scratch
- **Upcycling dense → MoE** beat continued dense training on equal budget

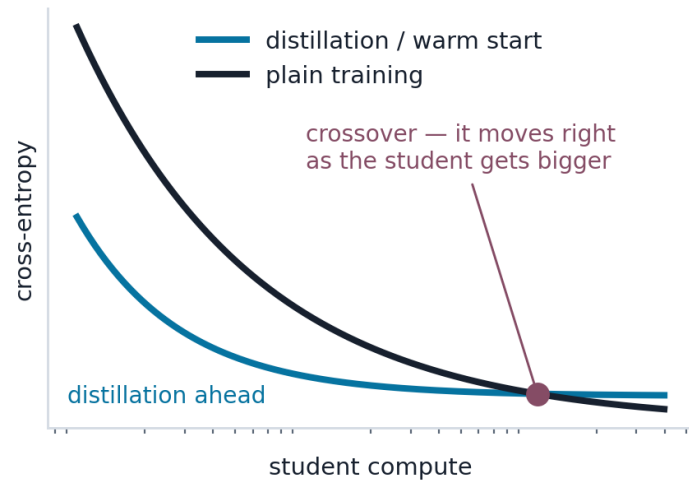


Minitron / Llama-3.1-Minitron, arXiv:2407.14679, 2408.11796 · Sparse Upcycling, arXiv:2212.05055



...and it sells the ceiling

- **Below a threshold** distillation wins — and the threshold grows with the student
- **Above it** plain training always catches up
- **At equal train loss** a warm-started network generalises worse



Curves schematic; the crossover is the result. Busbridge et al., Distillation Scaling Laws, arXiv:2502.08606 · Ash & Adams, On Warm-Starting Neural Network Training, NeurIPS 2020



Reuse builds families; frontiers restart

- **Between generations** the tokenizer, the context and the shape all change
- **Inherited pathologies** do not wash out with more tokens

Target	Start from
Flash / edge	your own Pro model
Domain model	open base + continual PT
New language	open base + continual PT
Pro / frontier	zero
Research claim	zero, matched budgets



What you tune, and in what order



01

Data

mixture and filters — the biggest lever

02

Architecture

size, sparsity, context — set by the target

03

Optimizer

and its stability tricks, on frozen data

04

Hyperparameters

LR, batch size, schedule — transferred, not searched

THREE RULES THAT COST MONEY WHEN BROKEN

No change ships without an ablation. Never swap framework between the ablations and the final run. Scale breaks what never broke small.



Next-token prediction is an exam in everything at once



Prefix	What predicting it requires
Having entered the room	grammar
Moscow is the capital of	world knowledge
$2x - 1 = 0; \quad x =$	arithmetic
<code>def fibonacci(</code>	code structure
«hello» in French is	translation
The witness lied, therefore	inference

WHY THIS OBJECTIVE WON

- **Every position** carries a loss — one pass, 5 signals
- **80–90% of documents** are under 2k tokens
- **So pack them** and mask across document borders

Without an intra-document mask a Python function attends to somebody's granola recipe. SmoLLM3 had the mask from day one; SmoLLM2 added it late.



03

• Data

What is in the web, what the pipeline does to it, and three results that go the wrong way



The most concentrated corpus is the curated one



- **A raw scrape is diverse** — no domain reaches one twentieth of a percent
- **Curation concentrates** — one preprint server is an eighth of RedPajama

Elazar et al., What's In My Big Data?, ICLR 2024 · arXiv:2310.20707, §4.2.2



Half of what you scrape, you already have

≈50%

duplicate documents

of RedPajama and LAION-2B-en, counted document for document

60–80%

removed by dedup

Common Crawl does not deduplicate across snapshots

800 GB

of RAM

what a naive exact-duplicate search over one corpus would need

The third number is why the next slides are about hashing. Elazar et al., ICLR 2024 · Common Crawl documentation



What you download is not the web

Format	What it holds	Used for
WARC	the full HTTP response, raw HTML	every serious pipeline
WAT	metadata, links, titles	graph and ranking work
WET	naive plain-text extraction	almost never in production



THREE FACTS THAT CHANGE WHAT YOU EXPECT

- **Sites block the crawler** — it is among the most blocked bots
- **Removals are contested** — coverage shifts under legal pressure
- **Ranking is harmonic centrality** — harder to game than PageRank



Extraction throws away exactly what you came for

raw HTML

```
<nav>Home Articles</nav>
<div id=cookie-banner>
  We use cookies.
  <article><h1>Attention</h1>
  <p>softmax(QKT/√d)V</p>
  <pre>def attn(q,k,v):</pre>
  <table>...</table>
```

naive WET

```
Home Articles
We use cookies. Accept
Subscribe - $1/month
Attention
softmax QK d V

def attn(q,k,v):
  | head | dim |
```

formula flattened, code and table gone

source-aware

```
Attention
softmax( $\frac{QK^T}{\sqrt{d}}$ )V

def attn(q,k,v):
  | head | dim |
```

head	dim
0	64
1	128

formula, code and table survive

- Formulas, code and tables die first — the worst loss for reasoning data
- So: **source-specific parsers** and model-based extraction where it pays

LANGUAGE ID

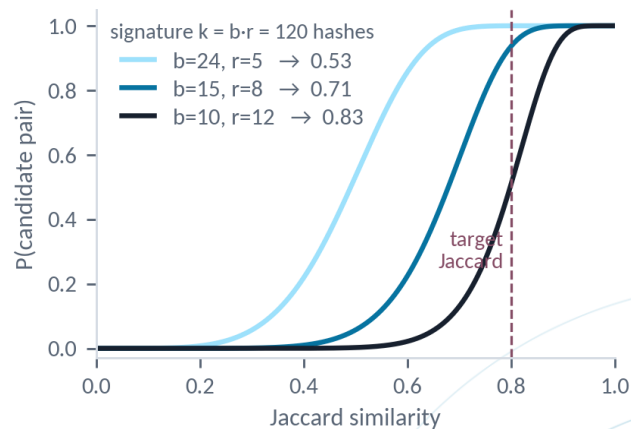
The threshold is a multilingual policy, not a constant.



A hash collision is as likely as the documents overlap

$$P[\min h(A) = \min h(B)] = |A \cap B| / |A \cup B|$$

- Shingle the document into overlapping w -word pieces
- Keep a signature of k min-hashes instead of the whole set
- LSH banding, $k = b \cdot r$ — compare only documents colliding in one whole band
- Cost drops from quadratic to nearly linear

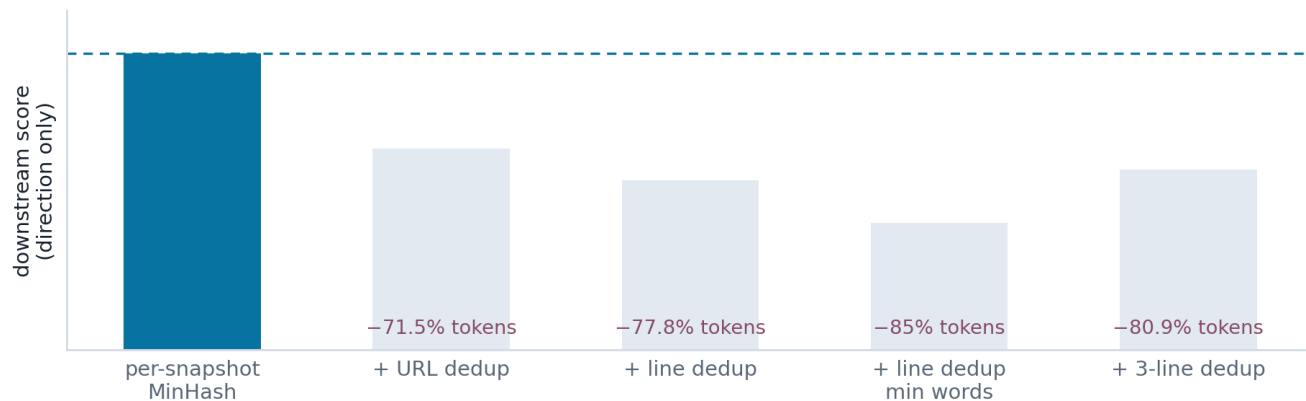


Three different letters: w -word shingles, a signature of $k = b \cdot r$ hashes, b bands of r . b and r decide where the curve turns — that is the knob you tune.



More deduplication made the model worse

Global dedup on top of per-snapshot MinHash: 71–85% more tokens removed, quality lost every time.



- **Why:** it up-samples what is unique, high-entropy and bad. Uniqueness is not value



Keep or drop?

Five documents from one crawl snapshot. Vote on each before the reveal.

A

Python file,
no prose

keep drop

B

Forum thread:
noise + one
good answer

keep drop

C

Conversational
Russian, slang,
no punctuation

keep drop

D

Well formatted,
confident,
factually wrong

keep drop

E

Product page:
spec table,
60 reviews

keep drop



The classifier gets four of the five wrong

A typical instruction-tuned quality filter drops A, B, C and E, and keeps D.

A DROP

code has no prose —
but code improves
reasoning

B DROP

noise ratio is high —
but the one answer
is the value

C DROP

wrong register —
and it is your
target language

D KEEP

looks like a textbook —
and it is false

E DROP

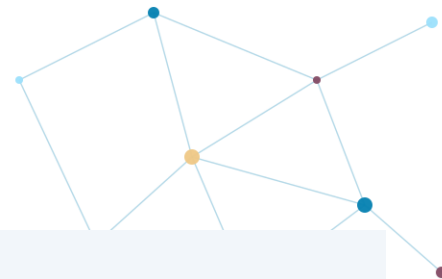
commercial boilerplate —
this one call
is right

AND THIS IS NOT A THRESHOLD PROBLEM

The filter scores form, not truth. No setting of it would have caught D.



The pipeline, in the order it runs



#	Stage	What it drops	Why it sits here
1	URL and robots filter	blocked domains, known spam	cheapest possible cut, before anything is parsed
2	Text extraction	navigation, banners, markup	everything downstream reads this text
3	Language ID	other languages	one cheap classifier, the single biggest cut
4	Exact dedup	byte-identical copies	a hash is free — do it before fuzzy
5	Fuzzy dedup · MinHash-LSH	near-copies	the big cut; its scope is a decision, not a default
6	Heuristic filters	junk recognisable by shape	cheap, so it runs before the model does
7	Quality classifier	low-value prose	expensive — it runs on the least data possible
8	PII and decontamination	personal data, test sets	applied to what will actually be trained
9	Mixing and shuffling	—	weights are chosen on the surviving pool



The cheap classifier beat the expensive one

TWO SCHOOLS

- **Label with a large model**, then distil into a light classifier
- **Or train cheap fastText** on an instruction-style positive set
- **The cheap one wins** — expensive labels ≠ a good boundary

WHAT RUNS BEFORE IT

- **Heuristics:** length, stop words, punctuation, repetition, caps
- **Compression ratio:** low means boilerplate, high means binary junk

after heuristics



after dedup



after the classifier



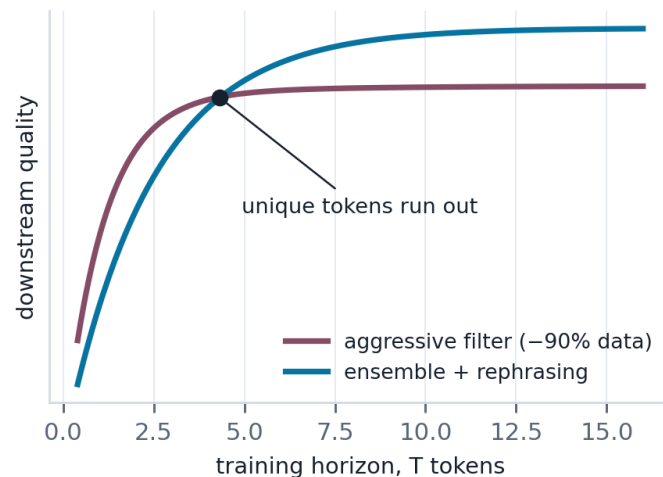
share of the original crawl still alive

DCLM-baseline, arXiv:2406.11794 — note DCLM runs heuristics before dedup, so these shares are not the order on the pipeline slide



Removing 90% of the data caps your horizon

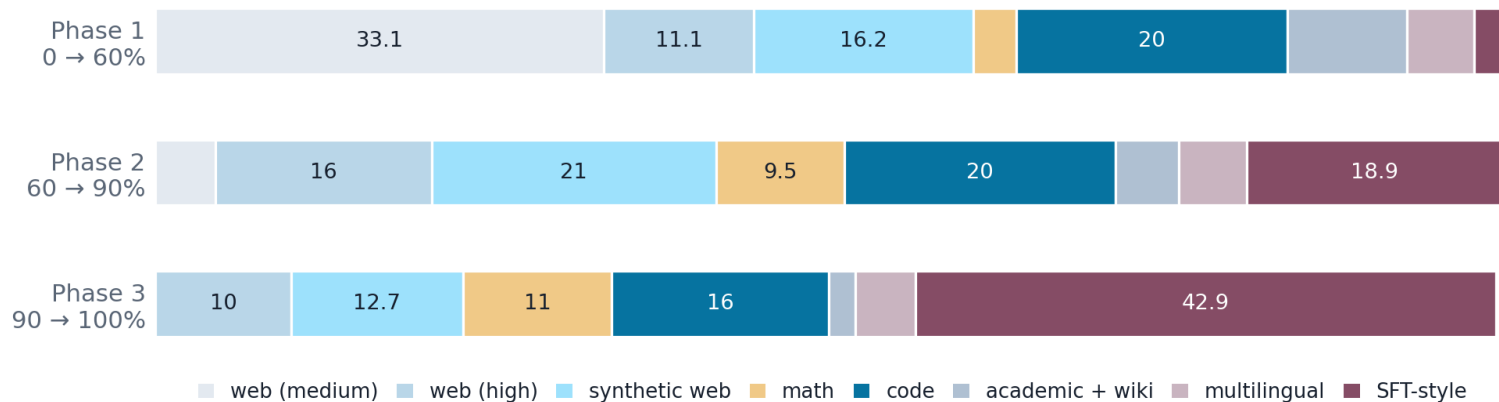
- **Aggressive filters win** on a 1T run and run out of unique tokens on 15T
- **The fix:** classifier ensemble, synthetic rephrasing, fewer heuristics
- **Result:** same quality on 4× more unique real tokens



Curves schematic; the numbers are Nemotron-CC, arXiv:2412.02595 — HQ subset +5.6 MMLU over DCLM at 1T tokens, full 6.3T corpus matches DCLM



The mixture is a hyperparameter, and it moves during the run

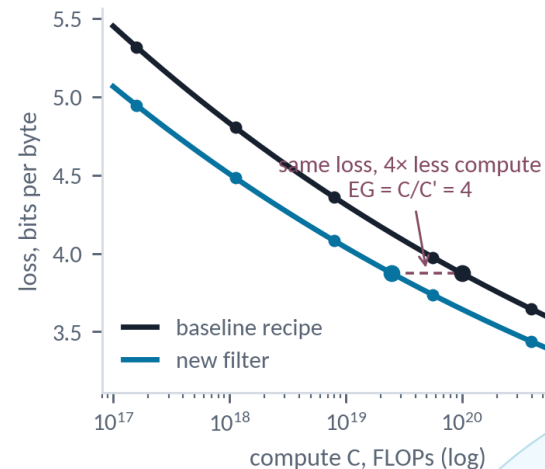


- **Web (medium) falls 33% → 0** while instruction-style data rises 3% → 43%
- **Automated by** group-DRO over domains, or regression over many small runs



None of this is visible from a single run

- **Ladder:** 3–5 sizes at a fixed tokens-per-parameter ratio
- **100–200 TPP for ablations**, 500–1000 for the final run
- **Fit $L = A \cdot C^\alpha + E$** , report efficiency gain $EG = C/C'$
- **Bits per byte** — perplexity is not comparable across tokenizers



A win measured at 100M parameters frequently disappears by 10B.



Four things to carry out of the data block

01 The web is a mall, not a library

half duplicates, benchmarks inside, PII everywhere

02 Every stage is a decision, not a default

extraction, language threshold, dedup scope, filter strength

03 Twice today, more was worse

extra dedup lost quality; 90% removal capped the horizon

04 Ladders, not points

fixed TPP, fitted curves, efficiency gain, BPB



04

• Optimization

Ten years of AdamW, and why the question reopened



AdamW: the W is the whole point

ADAMW — ONE STEP

$$m \leftarrow \beta_1 \cdot m + (1 - \beta_1) \cdot g$$

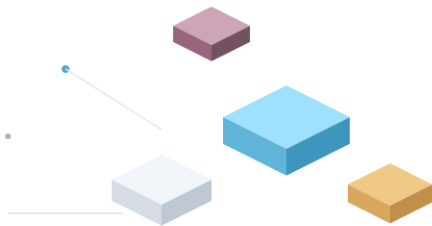
$$v \leftarrow \beta_2 \cdot v + (1 - \beta_2) \cdot g^2$$

$$\hat{m} = m / (1 - \beta_1^t), \quad \hat{v} = v / (1 - \beta_2^t)$$

$$\theta \leftarrow \theta - \eta \cdot (\hat{m} / (\sqrt{\hat{v}} + \epsilon) + \lambda \cdot \theta)$$

- m — direction, averaged over steps
- v — per-coordinate scale of the gradient
- $\lambda \cdot \theta$ sits outside the division by $\sqrt{v}$
- In **plain Adam** L2 goes through that division and gets distorted

Loshchilov & Hutter, Decoupled Weight Decay Regularization, ICLR 2019 · arXiv:1711.05101, Algorithm 2



Muon normalises directions, not coordinates

MUON — ONE STEP, PER MATRIX PARAMETER

$$B \leftarrow \mu \cdot B + G$$

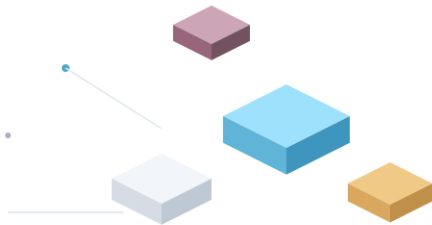
$$O \leftarrow NS_5(B) \approx U \cdot V^T$$

$$\theta \leftarrow \theta - \eta \cdot (0.2 \cdot \sqrt{\max(m, n)} \cdot O + \lambda \cdot \theta)$$

1-D params, embeddings, head $\rightarrow$ AdamW

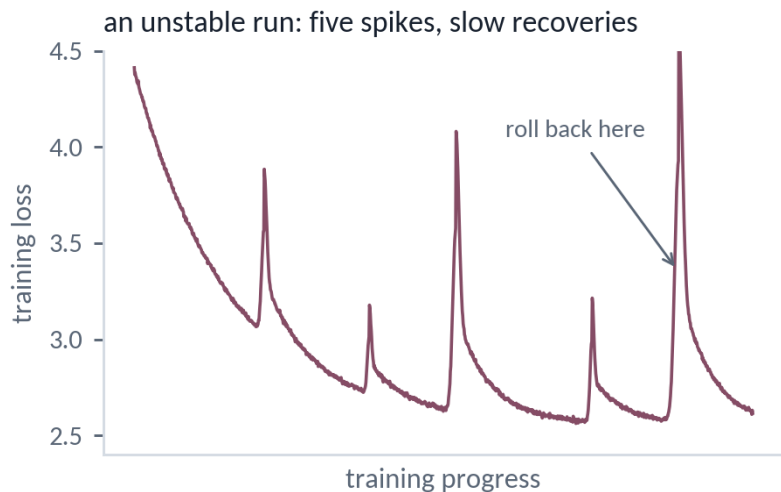
- NS_5 — five Newton–Schulz iterations in BF16, no SVD
- $U \cdot V^T$ is the momentum with every singular value set to 1
- **One buffer**, not two — adaptivity moved into the matrix op
- $0.2 \cdot \sqrt{\max(m, n)}$ matches AdamW's update RMS, so the LR transfers — and $\lambda \cdot \theta$ stays decoupled, exactly as on the previous slide

Jordan et al., Muon · Liu et al., Moonlight, arXiv:2502.16982, Algorithm 1



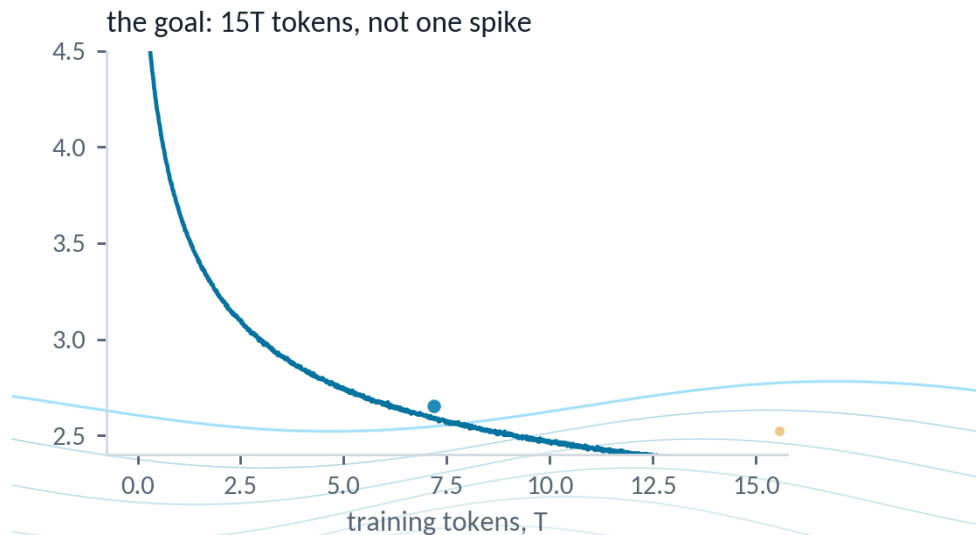
A spike is a genre, not an accident

- **Prevent:** global-norm clipping, warmup, z-loss, QK-norm
- **React:** roll back, skip the batch, then look at its data



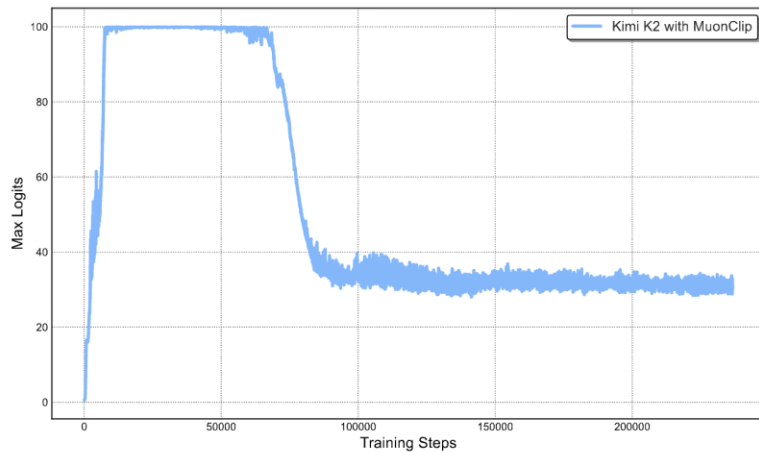
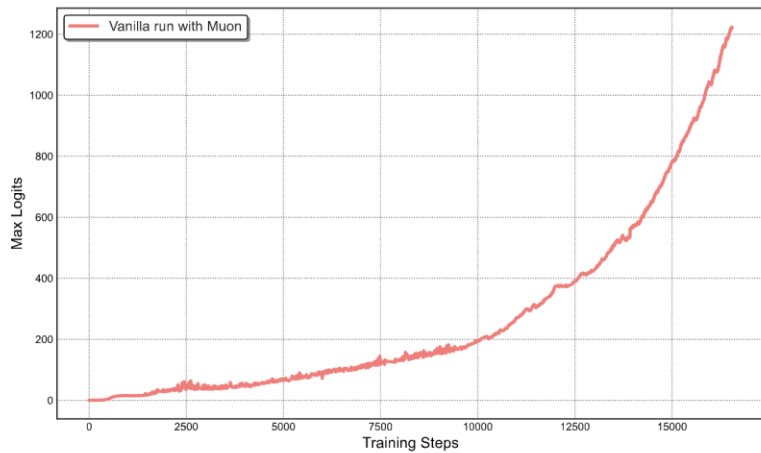
Curves schematic — the shape is the claim, not the numbers

- **The goal** is a curve with no spikes at trillion-token scale



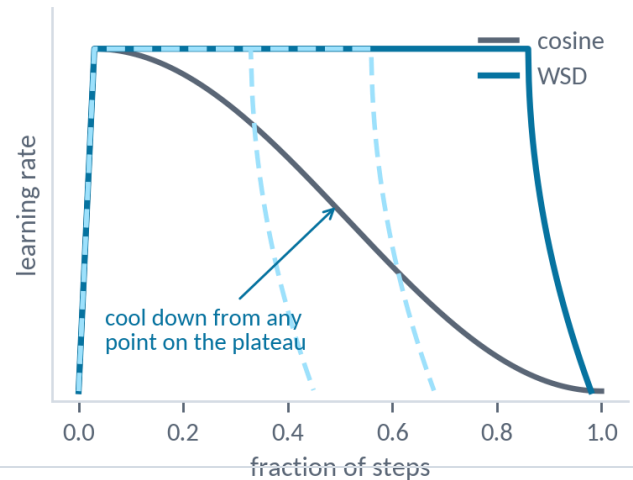
Orthogonalised updates blow up the attention logits

- **Reactive:** clip on top of the optimizer
- **Architectural:** normalise Q and K inside the block
- **In the rule:** orthogonalise per attention head
- **Honest status:** shipped by several labs; controlled gaps are smaller



WSD matches cosine — and unlocks everything else

- **Constant LR + cooldown** scales as predictably as cosine, at equal or better loss
- **Cosine forces** the step budget to be fixed in advance
- **WSD lets you** cool down from any point and reuse runs
- **Which is why** phases, mixture switches and continual PT are cheap

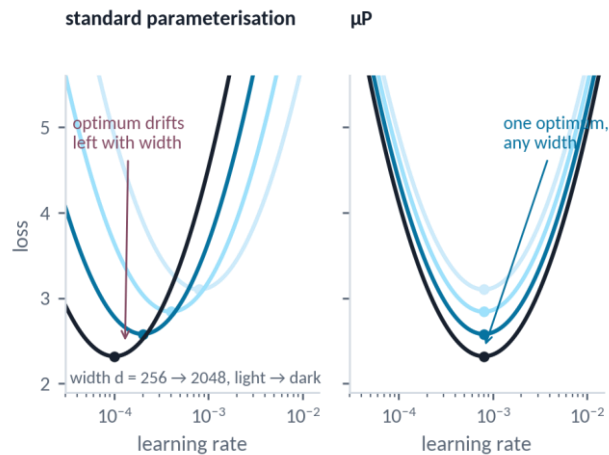


Hägele et al., Scaling Laws and Compute-Optimal Training Beyond Fixed Training Durations, NeurIPS 2024 · arXiv:2405.18392



The optimum moves — unless you fix the parameterisation

- **Standard:** the best LR drifts left every time the model gets wider
- **μP re-derives three things** so activations keep their size at any width: initialization scale, per-layer LR, and the output multiplier



Yang et al., Tensor Programs V / μ Transfer, arXiv:2203.03466



Step Law: formulas instead of a sweep

$$\eta(N, D) = 1.79 \cdot N^{-0.713} \cdot D^{0.307}$$

$$B(D) = 0.58 \cdot D^{0.571}$$

- **N** — number of parameters in the model
- **D** — number of training tokens

- The (LR, batch) surface is convex
- LR falls with model size
- Batch size depends on D only
- Fitted on a finite range

Predictable Scale / Step Law, arXiv:2503.04715 — constants fitted on dense models in a finite size range; outside it this is extrapolation



05

• Scale

How much model, how many tokens, in what order



$C \approx 6ND$, and it fits on a napkin

$$C \approx 6 \cdot N \cdot D$$

- **C** — total training compute, in FLOPs
- **N** — number of parameters in the model
- **D** — number of training tokens
- **6** — 2 forward + 4 backward, per parameter per token

SIXTY SECONDS, ON PAPER

1B parameters $\times$ 20B tokens = ?

At 40% MFU on eight accelerators of $1e15$ FLOP/s —
how long?

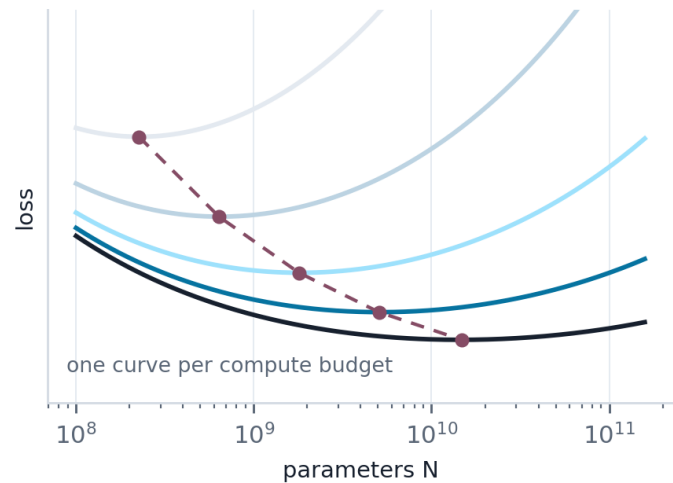
$1.2e20$ FLOPs $\div$ ($0.40 \times 1e15$ FLOP/s) $\approx 3.0e5$ s ≈ 83
accelerator-hours ≈ 10.4 h on eight.

Round numbers are the point: this is how a data decision
becomes a line in a budget.



GPT-3 was ten times undertrained

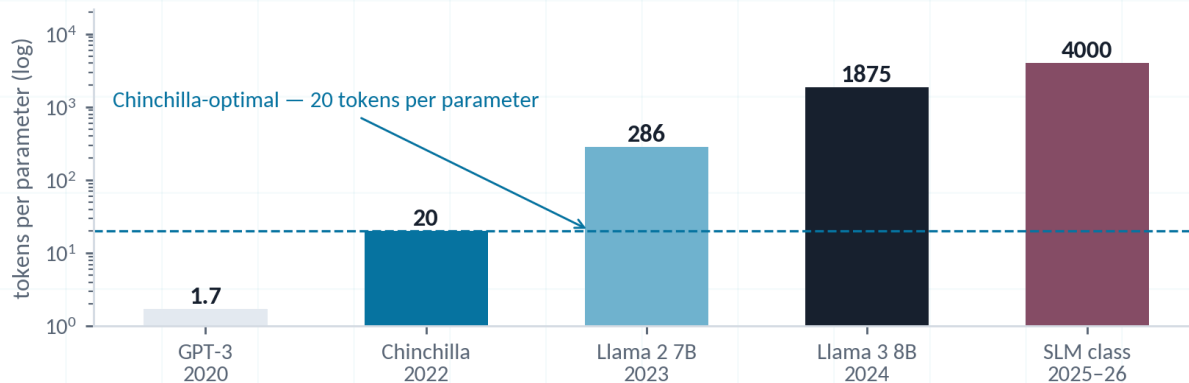
- **Kaplan, 2020:** grow the model — GPT-3 got 1.7 tokens per parameter
- **Chinchilla, 2022:** fix the budget, sweep N — the curve has a minimum near 20
- **2024:** the disagreement was explained, not waved away



Warmup, last-layer FLOPs and whether embeddings count in N · Kaplan, arXiv:2001.08361 · Hoffmann, arXiv:2203.15556 · Besiroglu, arXiv:2404.10102



Industry optimizes inference, not training

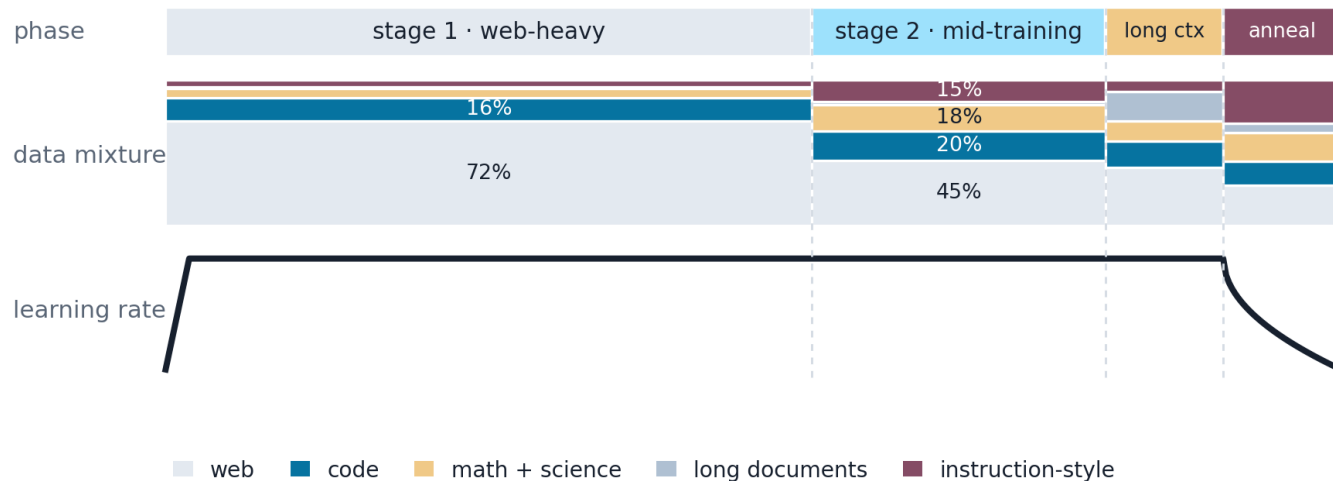


- **Inference-aware:** billions of requests ahead → over-train a smaller model
- **Data-constrained:** up to ~4 epochs ≈ fresh tokens; by ~16 the gain is gone
- **Sparsity:** at fixed active parameters, more experts keeps buying quality

Sardana et al., Beyond Chinchilla-Optimal, arXiv:2401.00448 · Muennighoff et al., Scaling Data-Constrained LMs, arXiv:2305.16264



A run is a schedule, not a curve



Three phases that are not warm-ups

Mid-training

- budget in trillions of tokens, not a tail
- explicit goal: a base for reasoning-RL
- instruction and math data rise to tens of percent

Long context

- natural long documents + synthetic long QA
- length raised progressively
- RoPE base frequency, interpolation, YaRN, NoPE

Continual PT

- naive restart gives a loss shock
- re-warm, re-decay, replay a few percent
- changes knowledge; SFT changes behaviour





Three times today, more turned out to be worse

more deduplication · more filtering · more model — each optimised a number that was not the goal



Eight things that were not obvious 90 minutes ago

- | | | | |
|----|--|----|--|
| 01 | The final run is a minority of the bill — 9.6–22.6% in the three disclosures we have | 05 | Curation concentrates a corpus far more than the raw web does |
| 02 | Experience transfers unevenly, and the catch-up effect is barely visible in data | 06 | Uniqueness is not value — extra dedup up-sampled high-entropy junk |
| 03 | The target class fixes size, sparsity, TPP and mixture before any ablation | 07 | A filter is only correct relative to a horizon, so a short run cannot judge it |
| 04 | A warm start buys tokens and sells the ceiling; frontiers restart | 08 | Ladders, fixed TPP, efficiency gain and BPB separate a result from an impression |



What to read and where to look



1 **Penedo et al.**

The FineWeb Datasets — the reference open ablation study
arXiv: 2406.17557

2 **Li et al.**

DataComp-LM (DCLM) — the cheap classifier that won
arXiv: 2406.11794

3 **Su et al.**

Nemotron-CC — the attack on aggressive filtering
arXiv: 2412.02595

4 **Elazar et al.**

What's In My Big Data? — duplicates, PII, contamination
ICLR 2024 · arXiv:2310.20707

5 **Epoch AI**

Final training runs are a minority of R&D compute
epoch.ai · 23.03.2026

6 **Hoffmann · Besiroglu**

Chinchilla, and the replication that reopened it
arXiv: 2203.15556 · 2404.10102

7 **Sardana · Muennighoff**

Inference-aware and data-constrained scaling
arXiv: 2401.00448 · 2305.16264

8 **Busbridge et al.**

Distillation Scaling Laws — when a warm start stops paying
arXiv: 2502.08606

9 **Hägele et al.**

Constant LR + cooldown scales like cosine
NeurIPS 2024 · arXiv: 2405.18392

10 **Ben Allal et al.**

The Smol Training Playbook — the messy version of this lecture
HuggingFace, 2025



Lecture 4: Pre-training II

efficient training

seminar

today, 15:30

materials

course repository, slides go up today

homework

two weeks, submission in the repo

next

GPU, memory, MFU, parallelism — the bill for today's recipe

